

Superconformal index and localizing higher derivative supergravity

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We show how equivariant localization can be used to compute the on shell action for supersymmetric $D = 5$ anti-de Sitter rotating, charged black holes in theories of supergravity with higher derivatives. An exact match with a dual field theory computation of the superconformal index in a Cardy-like limit is achieved.

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Introduction. The AdS/CFT correspondence provides key insights into the quantum properties of black holes and hence quantum gravity. A particularly interesting setting is supersymmetric anti-de Sitter AdS_5 solutions of string or M theory that are dual to $d = 4$, $\mathcal{N} = 1$ superconformal field theories (SCFTs). In this context, on the gravity side it is expected that the full supersymmetric partition function, with specific AdS_5 boundary conditions, should be equal to the superconformal index of the dual SCFT. In the large N limit the superconformal index has various complex saddles, which includes an important saddle that can be used to compute the black hole entropy after taking a Legendre transform. This saddle can be isolated by taking a certain Cardy-like limit and then taking the large N limit [1–5].

More precisely, consider a $D = 4$ $\mathcal{N} = 1$ SCFT on $S^1 \times S^3$, whose symmetry algebra contains J_1, J_2 generating the Cartan subalgebra of the $SO(4)$ symmetry of the round S^3 , and conserved charges Q_I ($I = 1, \dots, n + 1$) made of linear combinations of the superconformal R charge and n Abelian flavor charges. Choosing a complex supercharge $\mathcal{Q}$ satisfying $[J_1, \mathcal{Q}] = [J_2, \mathcal{Q}] = \frac{1}{2}\mathcal{Q}$, $[Q_I, \mathcal{Q}] = -r_I\mathcal{Q}$, the superconformal index is defined as the supersymmetric partition function on $S^1 \times S^3$,

$$\mathcal{I}(\omega_1, \omega_2, \varphi^I) = \text{Tr}[e^{-\beta\{\mathcal{Q}, \bar{\mathcal{Q}}\} + \omega_1 J_1 + \omega_2 J_2 + \varphi^I Q_I}], \quad (1)$$

where $\omega_1, \omega_2, \varphi^I$ are chemical potentials for the charges J_1, J_2, Q_I , respectively, subject to the constraint

$$\omega_1 + \omega_2 - 2r_I\varphi^I = \pm 2\pi i. \quad (2)$$

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Extending [1–6], it has been shown in [7] that the index in the Cardy-like limit $\omega_1, \omega_2 \rightarrow 0$, has the form

$$-\log \mathcal{I} = \frac{k_{IJK}\varphi^I\varphi^J\varphi^K}{6\omega_1\omega_2} - k_I\varphi^I \frac{\omega_1^2 + \omega_2^2 - 4\pi^2}{24\omega_1\omega_2}, \quad (3)$$

where $k_{IJK} = \text{Tr}(Q_I Q_J Q_K)$ and $k_I = \text{Tr}(Q_I)$ are cubic and linear 't Hooft anomaly coefficients, respectively. In (3) we have neglected terms that are exponentially suppressed in the Cardy-like limit, as well as a possible $\log |\mathcal{G}|$ term [8].

For holographic theories, the cubic anomaly term in (3) dominates at large N . Using two-derivative $D = 5$ supergravity, it has been shown in [9,10] that the on shell action for a complex locus of supersymmetric but nonextremal black holes, precisely agrees with the leading term in (3). Moreover, by taking a Legendre transform one obtains [11] the Bekenstein–Hawking entropy of the supersymmetric black holes of [12,13].

Recently there has been significant further progress in matching the subleading terms in (3) to a gravitational computation. Since we are considering subleading terms in N one necessarily needs to carry out the computation in the context of a $D = 5$ supergravity theory with higher derivatives. The result (3) depends on 't Hooft anomaly coefficients, and these are holographically determined from the Chern–Simons terms [14]

$$\frac{1}{24\pi^2} \left(k_{IJK} \mathcal{A}^I \wedge \mathcal{F}^J \wedge \mathcal{F}^K - \frac{1}{8} k_I \mathcal{A}^I \wedge \mathcal{R}_{ab} \wedge \mathcal{R}^{ab} \right), \quad (4)$$

where $\mathcal{A}^I$ are Abelian gauge fields dual to the R symmetry and flavor symmetry currents with $\mathcal{F}^I = d\mathcal{A}^I$, and $\mathcal{R}_{ab}$ is the Riemann curvature two form. One thus needs a $D = 5$ supergravity with four derivatives that provides a supersymmetrization of (4). The approach of [7,15–17] is to utilize conformal supergravity, which provides a powerful framework to deal with higher derivative theories [18]. A key ingredient is the identification of a supersymmetric

invariant, involving the supersymmetric completion of the square of the Weyl tensor, that contains the mixed gauge-gravitational Chern–Simons term [20]. Various off shell invariants contribute to the chiral anomaly; the proposal in [7] is to modify the two-derivative theory and combine it with an action built from the Weyl invariant.

This approach was used for minimal supergravity in [16,17] to match the universal part of (3), with flavor chemical potentials turned off. With vector multiplets, a matching was made in [7], focusing on equal angular momentum, certain theories, and also resorting to numerics. Given the topological nature and simplicity of the expression (3), one might expect [7] that it can be derived in a more robust and universal way using the techniques of equivariant localization in supergravity [21]. Indeed this has been achieved for the leading order term in [22–24].

The purpose of this paper is to show that localization can also be used to obtain the subleading terms. We utilize two ingredients: first, that localization of $D = 5$ supergravity at the two-derivative level can be carried out by performing a dimensional reduction and then utilizing results for $D = 4$, $\mathcal{N} = 2$ supergravity [24–26] and second, that localization can be extended to off shell conformal supergravity in $D = 4$ and hence to higher derivative theories [27]. The strategy is then clear: we consider the four-derivative supergravity theory in $D = 5$ of [7], dimensionally reduce to $D = 4$, and then use the results of [27]. We show that (3) can be recovered without requiring an explicit solution, without having to assume equal angular momenta, and for a general class of theories. Our procedure just assumes that a solution with certain prescribed global data exists [28].

Higher derivative supergravity. The $D = 5$ theory of [7] is constructed using off shell conformal supergravity, with suitable compensator multiplets and gauge fixing also required in order to recover Poincaré supergravity. The $D = 5$ superconformal multiplets consist of the Weyl multiplet coupled to n vector multiplets, one of which is a compensator, with an additional linear multiplet compensator, as in [31]. The two-derivative Lagrangian $\mathcal{L}_{2\partial}$ is controlled by a totally symmetric tensor C_{IJK} associated with the very special geometry of the scalar manifold. The effective off shell action considered in [7] has the form

$$\mathcal{L} = \mathcal{L}_{2\partial}|_{C_{IJK} \rightarrow C_{IJK} + \alpha \tilde{\lambda}_{IJK}} + \alpha \mathcal{L}_{\mathcal{C}^2}(\lambda_I), \quad (5)$$

where $\mathcal{L}_{\mathcal{C}^2}$ is the Weyl-squared Lagrangian [20] involving the vector multiplets, with λ_I a coupling, and one works to first order in the parameter α . The coupling $\tilde{\lambda}_{IJK}$ is a symmetric tensor which, in combination with C_{IJK} and λ_I , ensures that the theory can reproduce the anomalies of the dual field theory [7]. After gauge fixing we obtain parameters ζ_I that determine the gauging. To simplify subsequent formulas we set the gauge coupling $\mathfrak{g} = 1$.

By considering the resulting Chern-Simons terms that appear in the gauge-fixed action one can identify [7] the cubic and linear 't Hooft anomaly coefficients of the $D = 4$ SCFT with gravitational variables via

$$k_{IJK} = \frac{\pi}{4G_{(5)}} C_{IJK}^{(\alpha)}, \quad k_I = -\frac{24\pi}{G_{(5)}} \alpha \lambda_I, \quad (6)$$

where

$$C_{IJK}^{(\alpha)} \equiv C_{IJK} + \alpha \tilde{\lambda}_{IJK} - 8\alpha \lambda_{(I} \zeta_J \zeta_{K)}, \quad (7)$$

and $G_{(5)}$ is the $D = 5$ Newton constant [32]. Notice in particular that k_I is determined purely by the Weyl-squared Lagrangian. By contrast, k_{IJK} gets contributions from $\mathcal{L}_{2\partial}$, the Weyl-squared Lagrangian, and from other $D = 5$ invariants (e.g., see Table 1 of [33]), which are effectively incorporated together in $\tilde{\lambda}_{IJK}$. We also note that we have rescaled C_{IJK} in [7] via $C_{IJK} \rightarrow \frac{1}{6} C_{IJK}$ to agree with the notation of [24]; in particular, working to first order in α , the scalars Y^I in the $D = 5$ vector multiplets are constrained via $\frac{1}{6} C_{IJK} Y^I Y^J Y^K = 1$. Finally, as we will see, we can identify $\frac{1}{2} \zeta_I$ with r_I appearing in the $\mathcal{N} = 1$ algebra, $[Q_I, Q] = -r_I Q$.

Supersymmetric configurations of the $D = 5$ theory have a supersymmetric conformal Killing vector $\mathcal{K}$ that can be constructed as a bilinear in the Killing spinors. After gauge fixing this becomes a Killing vector. We are interested in solutions in which there is another Killing vector, ℓ , which is used to carry out a Kaluza-Klein (KK) dimensional reduction to $D = 4$, following [24]. We introduce a local coordinate so that we can write $\ell = \partial_{x^5}$ and introduce the KK ansatz for the metric $ds_{(5)}^2 = e^{-4\lambda} (dx^5 - A^0)^2 + e^{2\lambda} ds_{(4)}^2$. We can also write the $D = 5$ scalars and gauge fields in the form $Y^I = -e^{2\lambda} z_2^I$ (so $e^{6\lambda}$ is determined by the cubic constraint for Y^I), $A^I = A^I + z_1^I (dx^5 - A^0) + a^I dx^5$, where A^I are $D = 4$ gauge fields with $F^I = dA^I$. The a^I are constants which are defined up to some constant shift of the variables; quantities that are invariant are defined with a caron, e.g. $\check{z}_1^I = z_1^I + a^I$, $\check{A}^I = A^I + a^I A^0$, $\check{F}^I = F^I + a^I F^0$. The resulting $D = 4$ theory then has $n + 2$ vector multiplets, one of which can be viewed as a compensator multiplet, with $n + 2$ gauge fields $A^\Lambda \equiv (A^0, A^I)$ and $n + 1$ complex scalars $z^I \equiv z_1^I + i z_2^I$. The latter are associated with the constrained vector multiplet scalars $X^\Lambda = (X^0, X^I)$, where $z^I = X^I / X^0$.

By carrying out the dimensional reduction as discussed in [34], from (5) we obtain a $D = 4$ off shell $\mathcal{N} = 2$ conformal supergravity theory that is constructed from a Weyl multiplet, $n + 1$ vector multiplets, and a T-log multiplet. The resulting $D = 4$ theory can be expressed in terms of a prepotential $\mathcal{F}$, given by

$$\mathcal{F}(X^\Lambda, A_{\mathbb{W}^2}, A_{\mathbb{T}}) = (C_{IJK} + \alpha \tilde{\lambda}_{IJK}) \frac{\check{X}^I \check{X}^J \check{X}^K}{6X^0} + \alpha \lambda_I \frac{\check{X}^I}{X^0} (w A_{\mathbb{W}^2} + \mathfrak{t} A_{\mathbb{T}}), \quad (8)$$

where X^Λ , $A_{\mathbb{W}^2}$ and $A_{\mathbb{T}}$ are the lowest component fields of the vector, Weyl-squared, and T-log multiplets, respectively. The T-log multiplet is constructed from the vector multiplet containing X^0 [34], which is a constrained combination of a chiral and antichiral multiplet of weight $w' = 1$. Observe that $\mathcal{F}$ is homogeneous of degree 2 with respect to X^Λ , and degree 1 with respect to $A_{\mathbb{W}^2}$ and $A_{\mathbb{T}}$. In our normalizations (as in [27]), we have $w = \frac{1}{32}$ and $\mathfrak{t} = \frac{1}{3}$, but it is illuminating to keep w and $\mathfrak{t}$ to see how various terms contribute to the localization computation.

Localization. In [25,26] it was shown that the on shell action for Euclidean supersymmetric solutions to $D = 4$, $\mathcal{N} = 2$ gauged supergravity coupled to vector multiplets, at the level of two derivatives, localizes to the fixed points of the supersymmetric Killing vector ξ . Here ξ is constructed as a bilinear in the Killing spinors. Furthermore, all boundary terms in the action precisely cancel (after an appropriate Legendre transform ensuring supersymmetry), with the result expressed only in terms of the prepotential $\mathcal{F}$ evaluated at the fixed points, together with the weights of ξ and certain other topological data. Some key localization results of [25,26] have recently been extended to conformal supergravity constructions of general higher derivative theories in $D = 4$, including for prepotentials of the form (8) [27,35].

Thus, we can use the localization formulas of [25–27] to evaluate the $D = 5$ on shell action via the KK reduction described at the end of the last section. There are several subtleties. First, here we only consider solutions where we can assume the $D = 5$ gauge fields are globally defined one forms, which avoids issues with gauge dependence and an additional Chern-Simons-like contribution to the action. Second, we may circumvent the thorny issue of boundary terms and holographic renormalization in $D = 5$ by using a background subtraction prescription implemented in $D = 4$, as discussed in [24]. Third, the localization computation in [27] is for a $D = 4$ Euclidean theory with independent chiral and antichiral fields which were taken to be real. For our purposes, we need to consider a suitable analytic continuation which in practice means considering prepotentials $\mathcal{F}^\pm$ which are functions of the chiral and antichiral fields, which are no longer real, with $\mathcal{F}^\pm(X_\pm^\Lambda, A_{\mathbb{W}^2_\pm}, A_{\mathbb{T}_\pm}) \equiv \pm i \mathcal{F}(X_\pm^\Lambda, A_{\mathbb{W}^2_\pm}, A_{\mathbb{T}_\pm})$ [36].

The $D = 4$ supersymmetric conformal Killing vector ξ is simply the projection of the $D = 5$ vector $\mathcal{K}$ [24]. We denote the $D = 4$ spacetime by M_4 . We will only need to consider the case where ξ has isolated fixed points on M_4 , which are called “nuts” [37]. These are labeled $\text{nut}_\pm$, where

the $D = 4$ Killing spinor necessarily has either positive/negative chirality $\chi = \pm 1$ at each nut. The tangent space at such a point is $\mathbb{R}^4/\mathbb{Z}_d = (\mathbb{R}^2 \oplus \mathbb{R}^2)/\mathbb{Z}_d$, allowing for orbifold points of order $d \in \mathbb{N}$, and the weights of ξ on each of the $\mathbb{R}_i^2/\mathbb{Z}_d$ factors are denoted b_i , $i = 1, 2$, respectively. The arguments of $\mathcal{F}$ in (8) are rescaled by certain spinor bilinears $S_\pm$ in the localization formulas of [27], correlated with antichiral/chiral multiplets, respectively, in such a way that these also have simple localization properties at the fixed points. In particular, only $A_{\mathbb{W}^2_\mp}$, $A_{\mathbb{T}_\mp}$ contribute to the fixed point formula at a $\text{nut}_\pm$, respectively, via the key formulas [27]

$$S_\pm^2 A_{\mathbb{W}^2_\mp}|_{\text{nut}_\pm} = 16(b_1 \pm b_2)^2, \\ S_\pm^2 A_{\mathbb{T}_\mp}|_{\text{nut}_\pm} = \frac{w'}{2}(b_1 \mp b_2)^2, \quad (9)$$

and recall from above that $w' = 1$. Likewise, the following “dressed” versions of the scalars X^Λ (or their gauge-invariant versions $\check{X}^I \equiv X^I + a^I X^0$) appear in the fixed point formula

$$\Phi_0^\Lambda|_{\text{nut}_\pm} = \mp 2S_\pm X_{\mp}^\Lambda, \quad (10)$$

and we similarly define $\check{\Phi}_0^I \equiv \Phi_0^I + a^I \Phi_0^0$. The Φ_0^Λ in (10) are precisely the zero-form components of equivariantly closed forms associated to the gauge field curvatures:

$$d_\xi(F^\Lambda + \Phi_0^\Lambda) = 0, \quad (11)$$

with $d_\xi = d - \xi \lrcorner$ the equivariant exterior derivative, and we follow the normalizations in [27].

With all these definitions in hand, we can finally state the fixed point formula [27] for the $D = 5$ on shell action associated to the prepotential (8):

$$I_{4d}^{\text{FP}} = \frac{i\pi^2}{2G_{(5)}} \sum_{\text{nuts}_\pm} \frac{1}{d} \left\{ (C_{IJK} + \alpha \tilde{\lambda}_{IJK}) \frac{\check{\Phi}_0^I \check{\Phi}_0^J \check{\Phi}_0^K}{6b_1 b_2 \Phi_0^0} + 4\alpha \lambda_I \frac{\check{\Phi}_0^I}{b_1 b_2 \Phi_0^0} \left[16w(b_1 \pm b_2)^2 + \frac{1}{2}\mathfrak{t}(b_1 \mp b_2)^2 \right] \right\}, \quad (12)$$

where we have used (9). Here Φ_0^Λ , b_1 , b_2 and d depend on the nut; when we need to make this clear, we label the nuts by an index a and write $\Phi_0^\Lambda|_a$, b_1^a , b_2^a , d^a . The $\check{\Phi}_0^I$ are then subject to the constraint

$$-2Q^{(\ell)} \Phi_0^0|_a + \zeta_I \check{\Phi}_0^I|_a = \sigma_a(b_1^a - \chi_a b_2^a) \quad (13)$$

at each fixed point [24]. Here $\chi_a = \pm 1$ is the chirality of the a th $\text{nut}_\pm$, σ_a is another sign that we discuss momentarily,

and $Q^{(\ell)}$ is the charge of the $D = 5$ Killing spinor under the KK vector ℓ .

Finally, we will also use the following localization formula for the flux, via (11):

$$\frac{1}{4\pi} \int_D F^\Lambda = \frac{1}{2} \left(\frac{1}{d_N b_N} \Phi_0^\Lambda|_N + \frac{1}{d_S b_S} \Phi_0^\Lambda|_S \right), \quad (14)$$

where $D \subset M_4$ is a two-dimensional subspace to which ξ is tangent and nonzero. Necessarily then $D \cong \mathbb{WCP}^1_{[d_N, d_S]}$ is a weighted projective space/spindle, and the weights are $b_N = -b_0/d_N$, $b_S = b_0/d_S$ where $\xi|_D = b_0 \partial_\phi$ with ϕ having period 2π .

Black holes. We now evaluate the on shell action for a class of Euclidean black holes, using the formula (12). We regularize the action using background subtraction, subtracting the on shell action of Euclidean AdS_5 , as described in [24]. As advertised in the introduction, we do this without using any explicit black hole solutions; all we need is certain global data, which we now summarize, together with the assumption that such supergravity solutions exist.

The Euclidean black holes have topology $M_5 = \mathbb{R}^2 \times S^3$, with conformal boundary $\partial M_5 = S^1_\tau \times S^3$ where the superconformal index is defined. We introduce polar coordinates (r, τ) on the $\mathbb{R}^2$ factor, where τ parametrizes the thermal circle and has period 2π , while the horizon S^3 is at $r = 0$. The generators of the Cartan subgroup $U(1)^2 \subset SO(4)$ acting on S^3 are denoted ∂_{ϕ_i} , $i = 1, 2$, so that (τ, ϕ_1, ϕ_2) parametrize the torus $U(1)^3$. In this basis the $D = 5$ supersymmetric conformal Killing vector is

$$\mathcal{K} = \partial_\tau + \varepsilon_1 \partial_{\phi_1} + \varepsilon_2 \partial_{\phi_2} = (1, \varepsilon_1, \varepsilon_2), \quad (15)$$

where [24] defined $\mathcal{K}$ as a bilinear in the $D = 5$ Killing spinors. In (15), and subsequently, we identify vector fields with elements of the Lie algebra $\mathfrak{u}(1)^3 \cong \mathbb{R}^3$.

A special role is played by codimension two subspaces $\mathcal{D}_a \subset M_5$ that are fixed by $U(1) \subset U(1)^3$ subgroups. Writing the generating vector fields as $V_a \in \mathbb{Z}^3$, we have

$$\begin{aligned} \mathcal{D}_0 &\cong \mathbb{R}^2 \times S^1_1, & V_0 &\equiv (0, 0, 1), \\ \mathcal{D}_1 &\cong S^3, & V_1 &\equiv (1, 0, 0), \\ \mathcal{D}_2 &\cong \mathbb{R}^2 \times S^1_2, & V_2 &\equiv (0, 1, 0), \end{aligned} \quad (16)$$

where S^1_i is parametrized by ϕ_i . These describe the “rod structure” of [38], where we note that $\mathcal{D}_1$ is the horizon. Importantly, M_5 has no nontrivial two cycles, implying that the Abelian gauge fields $\mathcal{A}^I$ may be represented by global one forms. In this global gauge, the $D = 5$ Killing spinor ζ , which is charged under the combination $\zeta_I \mathcal{A}^I$, satisfies

$$\mathcal{L}_{V_a} \zeta = \frac{i}{2} \sigma_a \zeta. \quad (17)$$

Here $\sigma_a = \pm 1$ for each $a = 0, 1, 2$ simply follows from regularity of the spinor along each subspace $\mathcal{D}_a$.

The conformal boundary is $\partial M_5 = S^1_\tau \times S^3$, at constant radius $r \rightarrow \infty$. Euclidean global AdS_5 has the same conformal boundary, but different topology $N_5 = S^1_\tau \times \mathbb{R}^4$, where $\mathbb{R}^4$ fills the S^3 . Using this for background subtraction effectively leads [24] to the compact glued manifold $\bar{M}_5 \equiv M_5 \cup_{S^1_\tau \times S^3} (-N_5) \cong S^5$, which is topologically a five-sphere [39].

Reduction to $D = 4$. To reduce the black hole to $D = 4$ we need to pick a KK circle. The final $D = 5$ on shell action must be independent of this choice, and this is a nontrivial consistency check of our computation. We take the KK circle action to be an arbitrary $U(1)_\ell \subset U(1)^3$, specified by the primitive vector

$$\ell = (-p, n_1, n_2) \in \mathbb{Z}^3, \quad (18)$$

where $\gcd(p, n_1, n_2) = 1 = \gcd(n_1, n_2)$ [41]. From (17) we have $\mathcal{L}_\ell \zeta = iQ^{(\ell)} \zeta$, and hence we read off the charge

$$Q^{(\ell)} = \frac{1}{2} (-p\sigma_1 + n_1\sigma_2 + n_2\sigma_0). \quad (19)$$

The $D = 4$ spacetime is $M_4 = M_5/U(1)_\ell$, with projection $\pi: M_5 \rightarrow M_4$. This inherits a $U(1)^2 \cong U(1)^3/U(1)_\ell$ action from M_5 , which makes it toric. The $D = 4$ supersymmetric Killing vector is the projection of (15)

$$\xi \equiv \pi_* \mathcal{K}. \quad (20)$$

To apply the $D = 4$ localization formula (12) we need to compute the weights (b_1^a, b_2^a) of ξ at the fixed points $x_a \in M_4$. In M_5 , each such fixed point defines a circle fiber $S^1_a = \pi^{-1}(x_a)$, and we also define the weights w_a via

$$\mathcal{K}|_{S^1_a} = -w_a \ell|_{S^1_a}. \quad (21)$$

This determines the zero-form part of the equivariantly closed form associated to the KK gauge field [24],

$$\Phi^0|_a = w_a. \quad (22)$$

To compute all these data, we note that M_4 is topologically the total space of the complex line orbibundle $\mathcal{O}(-p) \rightarrow \mathbb{WCP}^1_{[n_1, n_2]}$ over a spindle. The glued manifold $S^5/U(1)_\ell = \bar{M}_5/U(1)_\ell \cong \mathbb{WCP}^2_{[p, n_1, n_2]}$ is a weighted projective space. In terms of weighted homogeneous coordinates (z_*, z_0, z_1) on this weighted projective space, the fixed points of $U(1)^2$ are $x_a = \{z_b = 0 | b \neq a\}$, for $a = 0, 1, *$, where x_* corresponds to the origin of

Euclidean global AdS_5 . These are orbifold points of order d^a , with $d^0 = n_1$, $d^1 = n_2$, $d^* = p$. The toric geometry of this example was studied in [42], and we may simply write down the weights

$$\begin{aligned} (b_1^0, b_2^0, w_0) &= \frac{1}{n_1} (n_2 \varepsilon_1 - n_1 \varepsilon_2, -n_1 - p \varepsilon_1, -\varepsilon_1), \\ (b_1^1, b_2^1, w_1) &= \frac{1}{n_2} (-n_2 - p \varepsilon_2, n_1 \varepsilon_2 - n_2 \varepsilon_1, -\varepsilon_2), \\ (b_1^*, b_2^*, w_*) &= \frac{1}{p} (-n_2 - p \varepsilon_2, -n_1 - p \varepsilon_1, 1). \end{aligned} \quad (23)$$

Next we turn to the fluxes. There is a single compact two cycle in M_4 , which is the projection of the black hole horizon $D_1 \equiv \mathcal{D}_1/U(1)_\ell \cong \mathbb{WCP}_{[n_1, n_2]}^1$. We define the gauge field fluxes through this cycle as

$$\begin{aligned} \mathfrak{p}^\Lambda &\equiv \frac{1}{4\pi} \int_{D_1} F^\Lambda \\ &= \frac{1}{2(n_2 \varepsilon_1 - n_1 \varepsilon_2)} (\Phi_0^\Lambda|_0 - \Phi_0^\Lambda|_1), \end{aligned} \quad (24)$$

where in the second equality we used the localization formula (14), together with the relevant weights in (23). Notice that using (22) and (23) the flux of the KK gauge field is $\mathfrak{p}^0 = -1/(2n_1 n_2)$, which is the correct Chern number for the fibration $S^3 \rightarrow \mathbb{WCP}_{[n_1, n_2]}^1$. Using the freedom to choose the flat connection in the KK reduction to $D = 4$, we now set

$$a^I = -\frac{\mathfrak{p}^I}{\mathfrak{p}^0} \Rightarrow \check{\mathfrak{p}}^I = 0 \Rightarrow \check{\Phi}_0^I|_0 = \check{\Phi}_0^I|_1. \quad (25)$$

On reducing the $D = 5$ Killing spinor to $D = 4$, the signs σ_a defined via (17) precisely become the signs defined in Ref. [42], and which appear in Eq. (13). The chirality of the $D = 4$ Killing spinor at the fixed point x_a is then χ_a , where $\chi_0 = -\sigma_0 \sigma_1$, $\chi_1 = -\sigma_1 \sigma_2$, $\chi_* = -\sigma_2 \sigma_0$ [42].

The constraint (13) is the same at all three fixed points x_0, x_1, x_* , and reads as

$$\sigma_1 + \sigma_2 \varepsilon_1 + \sigma_0 \varepsilon_2 = -\zeta_I \check{\Phi}_0^I|_{0,1,*}. \quad (26)$$

Indeed, necessarily

$$\check{\Phi}_0^I|_* = \check{\Phi}_0^I|_{0,1} \equiv \check{\Phi}^I \quad (27)$$

also holds. One can see this by appealing to the fact that via a UV/IR equation [24,27] the values $\check{\Phi}_0^I|_{0,1}$ on the horizon are related to quantities at the UV boundary, and that the same argument relates $\check{\Phi}_0^I|_*$ to the *same* boundary quantities. Alternatively, using the compact glued space $\tilde{M}_4 = \text{WCP}_{[p, n_1, n_2]}^2$ we have $H_2(\mathbb{WCP}_{[p, n_1, n_2]}^2, \mathbb{R}) = \mathbb{R}$ has only one independent two-cycle, and since $\check{\mathfrak{p}}^I = 0$ for the cycle

at $z_* = 0$ in (24), the same is true for the other two spindle cycles at $z_0 = 0$, $z_1 = 0$, which connect the poles of the horizon to the point x_* .

Substituting the above into (12), we find that all dependence on ℓ in (18) indeed cancels out, giving

$$\begin{aligned} I_{40}^{\text{FP}} &= \frac{i\pi^2}{2G_{(5)}} \left\{ -(C_{IJK} + \alpha \tilde{\lambda}_{IJK}) \frac{\check{\Phi}^I \check{\Phi}^J \check{\Phi}^K}{6\varepsilon_1 \varepsilon_2} \right. \\ &\quad + 4\alpha \lambda_I \check{\Phi}^I \left[\left(16w - \frac{1}{2} \mathfrak{t} \right) \frac{(\sigma_1 + \sigma_2 \varepsilon_1 + \sigma_0 \varepsilon_2)^2}{\varepsilon_1 \varepsilon_2} \right. \\ &\quad \left. \left. - 32w \left(\frac{\varepsilon_1^2 + \varepsilon_2^2 + 1}{\varepsilon_1 \varepsilon_2} \right) \right] \right\}. \end{aligned} \quad (28)$$

Using the constraint (26), the second line may be combined with the first line. Setting $w = \frac{1}{32}$, $\mathfrak{t} = \frac{1}{3}$, the values obtained from dimensional reduction of the $D = 5$ action (5), remarkably we find this precisely combines into $C_{IJK}^{(a)}$ given in (7). The upshot is

$$\begin{aligned} I_{40}^{\text{FP}} &= -\frac{i\pi^2}{12G_{(5)}} C_{IJK}^{(a)} \frac{\check{\Phi}^I \check{\Phi}^J \check{\Phi}^K}{\varepsilon_1 \varepsilon_2} \\ &\quad - \frac{2i\pi^2}{G_{(5)}} \alpha \lambda_I \check{\Phi}^I \left(\frac{\varepsilon_1^2 + \varepsilon_2^2 + 1}{\varepsilon_1 \varepsilon_2} \right). \end{aligned} \quad (29)$$

This result and the constraint (26) can be precisely matched with the Cardy-like limit of the superconformal index (3) and (2), respectively. To do so we use the change of variables (6), giving the map between 't Hooft anomaly coefficients and gravity variables, together with

$$\omega_i \leftrightarrow 2\pi i \varepsilon_i, \quad \varphi^I \leftrightarrow -2\pi i \check{\Phi}^I, \quad r_I \leftrightarrow \frac{1}{2} \zeta_I. \quad (30)$$

To match, we have taken $\sigma_0 = \sigma_2 = +1$ which, from above, corresponds to the $D = 4$ Killing spinor having the same chirality at the poles of $\mathbb{WCP}_{[n_1, n_2]}^1$ (i.e., a topological twist), with $-\sigma_1$ specifying minus the charge of the $D = 5$ Killing spinor around the thermal circle, matching the $\pm$ sign on the right-hand side of (2) [43].

Discussion. We have obtained a universal derivation of the Cardy-like limit of the superconformal index using localization in supergravity. Our procedure uses a background subtraction prescription that attached a Euclidean AdS_5 factor at the boundary which effectively leads to implementing localization on a compact manifold without boundary. Interestingly, the final answer gets a contribution from three fixed points after dimensional reduction, two arising from the black hole horizon and another from the center of the AdS_5 subtraction factor [44]. It would be desirable to obtain our result using holographic renormalization rather than background subtraction, but there is still much to be understood regarding $D = 5$ holographic renormalization before this can be attempted.

For the black hole solutions we were able to assume that the $D = 5$ gauge fields are globally defined one forms. In order to generalize our results to compute higher derivative corrections to the on shell actions for more general supersymmetric solutions, including black rings and lenses (for related discussions at the two-derivative level, see Refs. [23,38,46]) we need to relax this assumption [47]. Finally, in this paper we have focused on gauged supergravity, but our key results also apply to ungauged supergravity with additional applications. We aim to report on these topics soon.

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